

Artem Kustikov

Principal Consultant, AI/DevOps/FullStack Software Developer



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Principal consultant and hands-on software architect with 20+ years of experience designing and delivering scalable, cloud-native systems end to end. I lead architecture and engineering across **AI/LLM-powered applications, distributed event-driven platforms, and full-stack web products**, translating business goals into resilient, well-governed solutions. Core focus areas: GenAI and agentic systems (LLM agents, evaluation and observability, MCP), solution and microservices architecture, cloud infrastructure (AWS, Azure, GCP), DevOps and platform engineering (Kubernetes, Terraform, CI/CD), and real-time data streaming (Kafka, Flink). I pair technical leadership with deep delivery experience — mentoring teams, driving engineering best practices, and shipping production systems that scale.

Open Source Projects

- **KateChat** — Self-hosted, multi-provider LLM chat platform (open ChatGPT alternative). React/TypeScript frontend with Node.js and Rust backends; integrates AWS Bedrock, OpenAI, and Yandex AI with RAG (Docling), MCP tool servers, in-browser Python (Pyodide), and image generation. GraphQL API, WebSocket subscriptions, PostgreSQL/Redis, Docker.
- **docling.rs** — High-performance Rust reimplementation of Python Docling, adopted into the official Docling project, that converts 20+ document formats (PDF, DOCX, PPTX, XLSX, HTML, EPUB, images, audio) into a unified `DoclingDocument` for AI/RAG pipelines. Pure-Rust PDF parser with an ONNX layout/TableFormer/OCR stack and Whisper ASR; 2–57× less memory and up to 46× faster than Python, with Node.js/TypeScript bindings. Emits Markdown, docling-JSON and DocLang (`.doclx`) conformant with upstream docling, and ships a pluggable RAG subsystem — chunking, swappable embedders and vector stores, hybrid/multi-query/HyDE retrieval, and an API-key-protected REST service.

Experience

Aug 2022 - Current

Principal Consultant, AI/DevOps/FullStack Software Developer

[Machine Learning Reply](#) Vienna, Austria

Stack: Node.js, React, Next.js • Java, Quarkus, Flyway • Python, PydanticAI • PostgreSQL, Cosmos DB, Redis • Apache Kafka, Flink, Confluent • Azure OpenAI, AWS Bedrock, Langfuse, MCP • Terraform, Kubernetes, Helm • AWS, Azure

Solution Architect/AI Engineer - *Self-service agentic AI platform*

- Owned end to end a self-service agentic platform where a non-engineer describes an AI agent in chat, gets it back as an editable visual workflow, publishes it, and real inbound traffic runs it over a shared mailbox, embeddable web forms, live phone calls (WebRTC with short-lived ephemeral tokens) and cron schedules; agents ground answers in Azure AI Foundry vector stores, reply in the customer's language, hand results to Salesforce and SAP, and reach customer systems through OAuth2-secured MCP servers with tokens encrypted at rest.
- Designed the architecture: a Next.js portal, two Python Azure Function Apps (dispatcher and executor on PydanticAI over the Azure OpenAI Responses API) and Storage Queues as the only seams between them, tied together by one workflow JSON contract shared by the ReactFlow canvas, the generation bot and the runtime — with streamed output, retry classification and a Redis-backed human-in-the-loop pause/resume primitive.
- Made it multi-tenant by construction (domain-based tenant resolution, a tenant predicate on every query, a build-failing test for any query that omits it) and shippable either as a shared portal with per-customer branding or as a single-tenant install in the customer's own Azure subscription; Terraform and GitHub Actions with OIDC stand it up from an empty subscription, and onboarding a customer is a GitHub Environment plus a tfvars file.

AI/ML Engineer - *GenAI demonstrators for sales and offer processes*

- Built an LLM-as-a-Judge quality layer for customer-facing chatbots: every conversation turn is traced to Langfuse and scored asynchronously by an AWS Bedrock LLM judge combined with coded metrics, surfacing 9 live quality scores (quality, cost, escalation risk) — designed to plug into any chatbot.

Solution Architect/FullStack - *Scheduling, billing and contract platform for concert organisers*

- Designed and built a multi-tenant platform serving several legally separate companies and event formats: one login supports multiple roles with role-specific features, and every query is scoped through explicit per-company access grants that can be gated on a signed contract.
- Modelled recurring schedules that generate concerts with immutable composition snapshots; double-booking is prevented in the database itself. Added . ICS confirmation emails with confirm/cancel deep links and PDF participant and fee reports.
- End-to-end TypeScript on Bun — GraphQL Yoga, Prisma/PostgreSQL, React/Redux/Mantine, Google and Apple OAuth — on AWS ECS Fargate with Terraform/Terragrunt and GitHub Actions; built end to end with Claude Code, from requirements and data model to infrastructure, as a production-scale AI-assisted development case study.

Senior DevOps/FullStack - *Web-client for artificial intelligence chatbot system*

- Develop robust and resilient CI/CD platform with integrated unit and e2e tests (Playwright) and support for feature environments.
- Introduce seamless user authentication and role-based authorization against AWS Cognito.
- Work on support for collaborative chats, workspaces, and documents based on WebSockets and WebRTC.

Senior Java Developer/DevOps - *Unified vehicles data streaming platform*

- Designed and implemented the Quarkus platform services — the Management API for schemas, data orders, pipelines and provisioning, and the Deployment Service provisioning Kafka topics, ACLs and identity pools on Confluent — integrated with Schema Registry and the customer's legacy vehicle-data services, with schema validation on ingestion and Kafka key selection fixed for ordering guarantees.
- Owned the Flink layer: pipeline persistence on reactive Hibernate/Flyway with automatic restart of failed jobs, a byte-for-byte refactor of the core job, a Redis-backed rules cache, the Flink 2.2 upgrade, and the migration tooling (export, rewrite, import, resumable batches) plus Cilium/Kyverno/ArgoCD/Helm codification for the move to the new Kubernetes platform.
- Hardened and scaled the Azure edge: mTLS for consumers, in-app JWT validation behind an off/shadow/enforce switch, managed identity instead of SPN credentials, and API Management policies with a shared Redis token cache (99% hit rate) that removed ~4.9M redundant calls per 12 h, lifted the 31-unit capacity ceiling and cut Log Analytics ingestion by ~EUR 43k/month.

DevOps Engineer/Senior Developer - *Online Sales Forecasting Tool*

- Refactor existing API and ML inference microservices written on Java (Spring Boot), Python (FastAPI), and R (plumber) to support deployment into k8s. Optimize Docker images build and versioning, add health checks and external configuration. Develop CI/CD framework based on Github actions. Implement custom Github action to perform HTTP polling.
- Migrate legacy infrastructure from AWS (RDS and EC2 managed with CloudFormation) to private OpenShift cluster. Setup continuous deployment system based on Helm charts/templates and Tekton triggers and pipelines.

May 2018 - Feb 2022

System Architect/Senior FullStack Software Developer

Intetics Minsk, Belarus

Stack: Node.js (express, restify), ReactJS (TypeScript, redux, lerna, grpc-web), Go lang (GRPC, protobuf), Threedium, Rust, Docker, Kubernetes, Google Cloud Platform, MongoDB, PostgreSQL, Bigtable

Worked on a large B2B project in the fashion industry, evaluated as #1 in the US market. Participated in various integration tasks and custom ETL (extract, transform, load) engine development. Worked on migration of a legacy frontend application from an Ampersand.JS-based framework to ReactJS.

- Profiled and optimised Node.js microservices; developed a rich interactive UI for the ETL tool with [React Flow](#), [dagre](#), and GRPC.
- Designed and implemented [Box.com](#) and [Dropbox](#) connectors for the ETL engine (Go lang); integrated the excelize library into its XSL processor and upstreamed several [fixes](#) to the library.

- Integrated [Threedium](#) 3D models with a custom React component.

Oct 2008 - May 2018

System Architect/Senior Software Developer

[EffectiveSoft](#) Minsk, Belarus

Participated in 20+ projects including NLP and text mining tool Intellexer.

Stack: Node.js (express), React, Angular, Ext.js, AWS (EC2, ElasticBeanstalk, RDS, CloudFront), MySQL, Redis, MongoDB, .NET (C#/Managed C++), ASP.NET MVC, C++, COM, WinAPI, ActiveMQ, Python.

- **Crowdfunding software** — B2B platform for local businesses with tight integration of the [Dwolla](#) payments service: web client and admin app architecture, CI setup (Bitbucket Pipelines, AWS CloudFormation), and a Node.js background worker for money transfers, interest charges, and financial audit.
- **Medicine: DICOM/ECG parsing and analysis** — universal ECG data loader replacing duplicate per-format implementations (Physionet, EFS, ISHNE, HL7), a canvas-based web client to upload and view DICOM images (zoom, WL-transformation, interactive measuring), and an ASP.NET MVC/SignalR clinic personnel synchronization tool.

Oct 2006 - Oct 2008

Senior Software Developer

InventionMachine/[IHS Markit](#) Minsk, Belarus

Stack: .NET, C#, C++, ATL/MFC, JavaScript/AJAX, Java, ColdFusion

Jun 2004 - Sep 2006

Software Developer

[SCAND](#) Minsk, Belarus

Stack: ASP 3.0 (VBScript), JavaScript/AJAX, C#, C++/boost/pthread, Java/Spring, MS SQL Server, Oracle

Dec 2002 - Jun 2004

Postgraduate student, teacher

Belarusian National Technical University Minsk, Belarus

Worked on robot simulation and analytical programming system – complex computer-aided modeling system for simulation of real industrial robots and their environment, calculation of robot kinematics, collision detection and analytical programming. Also worked as teacher in the staff of BNTU and taught several IT-related courses: Computer Networking Fundamentals, Mathematical Fundamentals of Programming of Robots.

Skills

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- **JavaScript Full Stack:** (2002-now) TypeScript, React/redux, Angular, Node.js, Express, Next.js, REST/GraphQL
 - **DevOps:** (2015-now) Docker, Terraform, Kubernetes, AWS, Azure, GCP, Gitlab
 - **Python:** (2008-now) Django, Flask, FastAPI, SQLAlchemy, Celery, NumPy, Pandas, nltk, Seaborn, Pytorch, scikit-learn
 - **Java:** (2004-now) Java 1.3/21, Struts/FOP/JSTL/POI, JBoss, Tomcat, Spring Framework, Spring Boot, Quarkus, Gradle
 - Performed 500+ technical interviews in JavaScript, DevOps, Java, .NET, and Golang

Education

Belarusian National Technical University | Minsk, Belarus 1997-2002 | Computer Science, Robotics

Certifications

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- Jun 2026: [Microsoft AI & ML Engineering](#)
 - Jan 2026: [AWS Generative AI Applications](#)
 - Oct 2025: [Confluent Certified Developer for Apache Kafka](#)
 - May 2024: HashiCorp Certified: [Terraform Associate \(003\)](#)
 - Oct 2023: [AWS Certified Solutions Architect – Associate](#)
 - Aug 2022: [DevOps on AWS](#)
 - Apr 2017: [Machine Learning and Data Analysis from MIPT/Yandex](#)
 - Sep 2016: [Microsoft Certified Solutions Associate: Web Applications](#)
 - Feb 2013: Microsoft Certified Solutions Developer: Web Applications

Languages

Russian (native) • English (full professional), IELTS 6.5, CEFR B2 • German (professional), OIF Integrationsprüfung B1